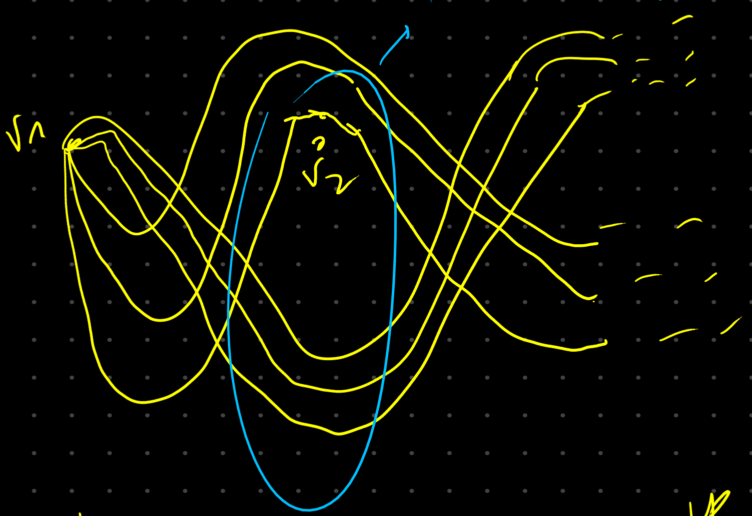
< Special cases from Hanani-Tutte to x-monotone drawing

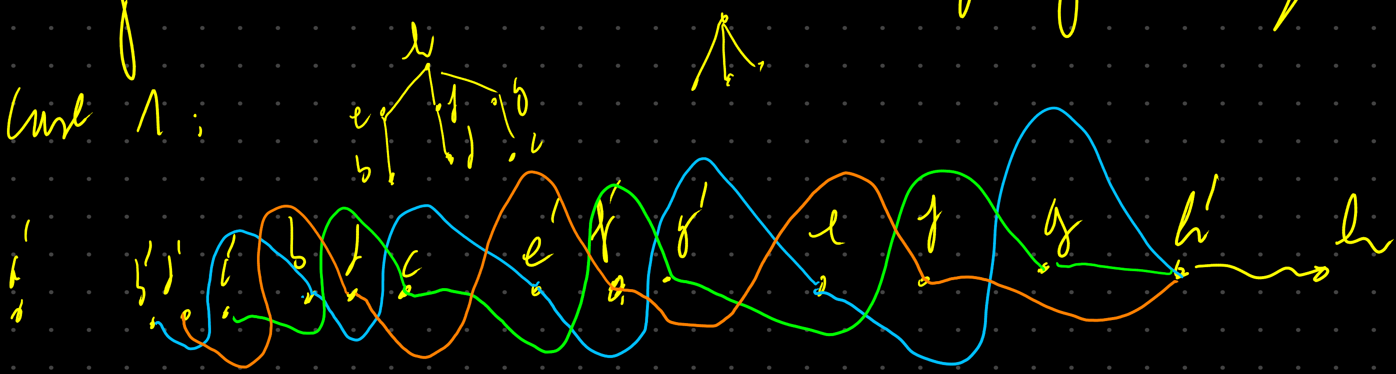
* Weak H-T: multiple connected comps linked by vertex v_1

this case cannot happen if $G \setminus \{v_1\}$ is connected.

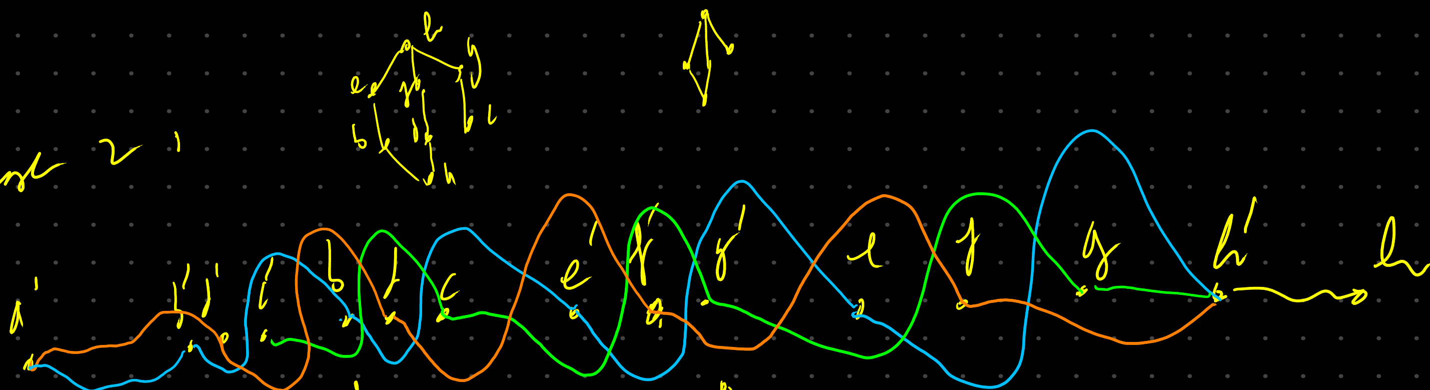


* Strong H-T: issues with odd crossings of adjacent edges.

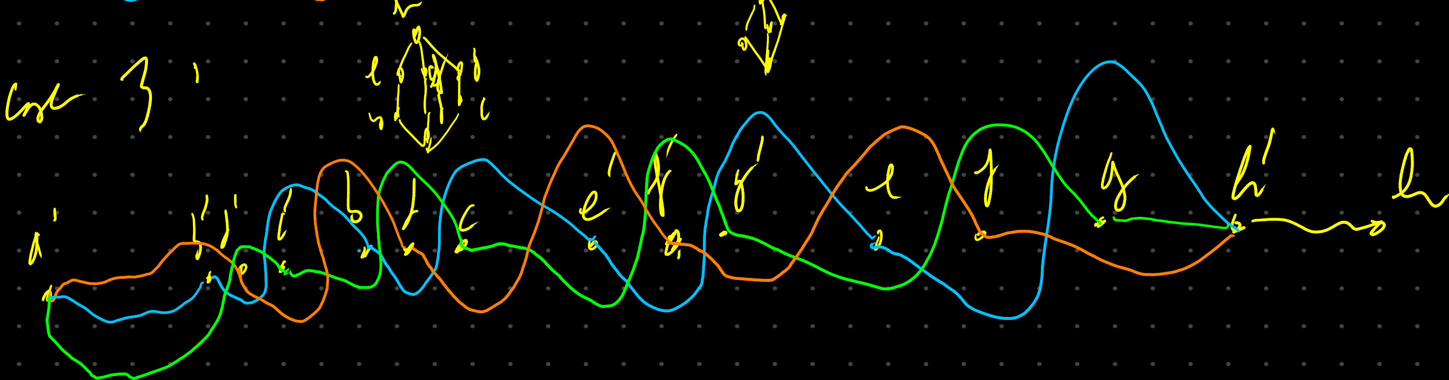
Case 1:

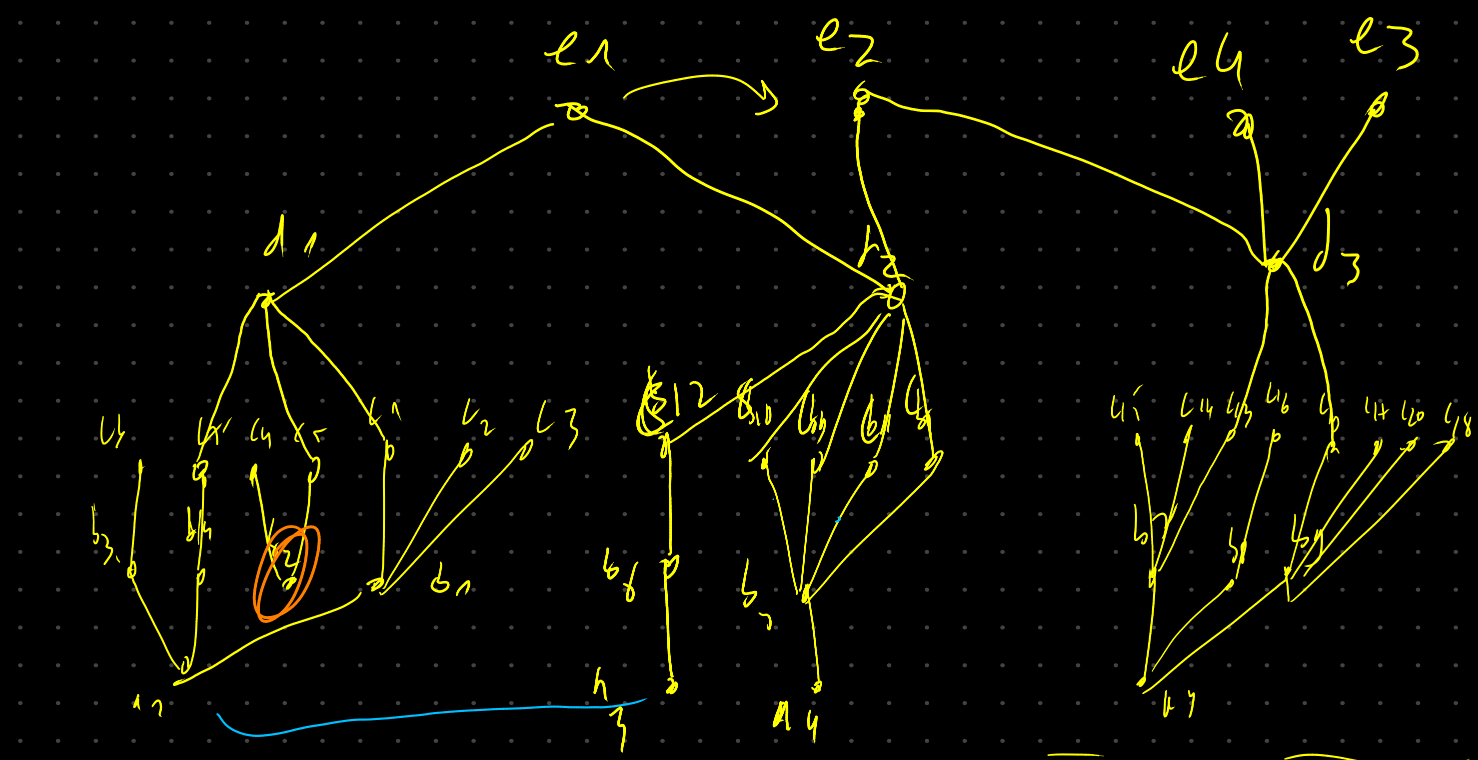


Case 2:

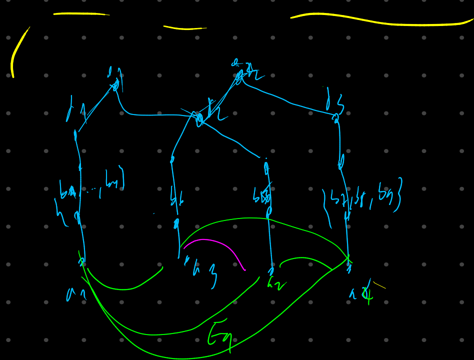


Case 3:





This graph contains all the cases from H-T drawing that might lead to a circular relation (we consider h to be connected)



simplify graph to find level 1 Equivalent classes

Equivalence Classes representation Matrices:

level 2

	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8	b_9
b_1									
b_2	E_{12}								
b_3	E_{13}	E_{15}							
b_4	E_{14}	E_{16}	E_{17}						
b_5	E_1	E	E_1	E_1					
b_6	E_1	E_1	E_1	E_1	E_2				
b_7	E_1	E_1	E_1	E_1	E_1	E_1			
b_8	E_1	E_1	E_1	E_1	E_1	E_1	E_{41}		
b_9	E_1	E_1	E_1	E_1	E_1	E_1	E_{42}	E_{43}	

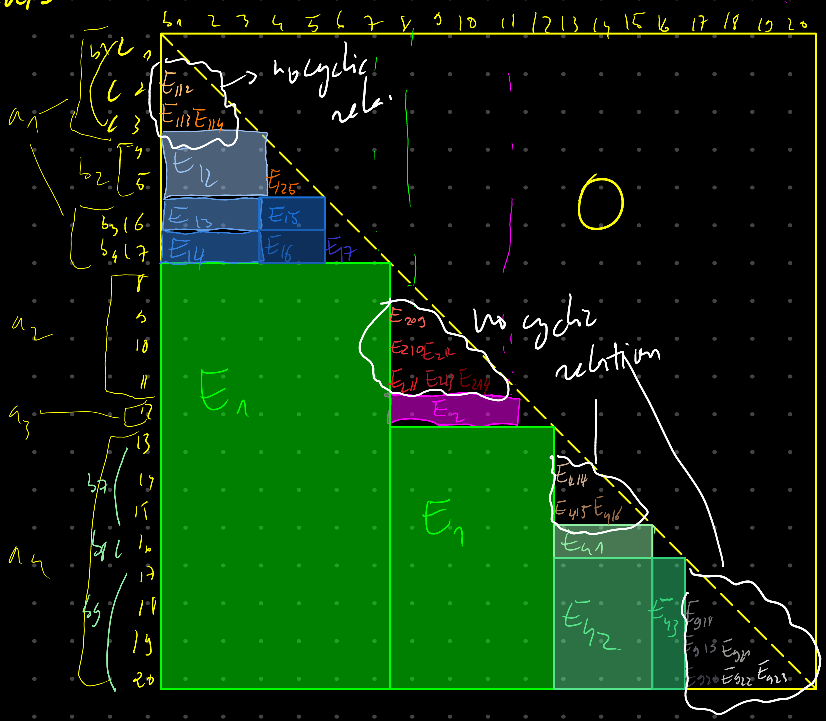
no cyclic relation

no cyclic relation

	a_1	a_2	a_3	a_4
a_1				
a_2	E_1			
a_3	E_1	E_2		
a_4	E_1	E_1	E_1	

level 1

Level 3



level 4

$$\begin{array}{c|ccc} & l_1 & l_2 & l_3 \\ \hline l_1 & & & \\ l_2 & E_1 & & O \\ l_3 & E_1 & E_1 & \end{array}$$

level 5

$$\begin{array}{c|cccc} & e_1 & e_2 & e_3 & e_4 \\ \hline e_1 & & & & \\ e_2 & E_1 & & O & \\ e_3 & E_1 & E_1 & & \\ e_4 & E_1 & E_1 & E_0 & \end{array}$$

- find vertices v with no incoming edges and $\exists\, w, w' \in V(h)$ with $\ell(w) = \ell(w') = \ell(v)$ and $\exists\, x \in V(h)$ with $(x,w), (x,w') \in E(h)$

v w x — $\{v\}$ is problematic

We need to specify same relative order

$\forall\, w, w' \in V(h)\ \exists\, x \in V(h)$ s.t.

$$\begin{cases} (x,w), (x,w') \in E(h) \\ \ell(w) = \ell(w') = \ell(v) \end{cases}$$

then

$$f(w,v) = f(w',v)$$

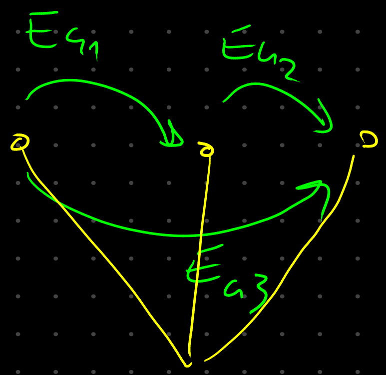
we can fill all the eq. c sles

that can lead to a cyclic relation

by making them equal each other

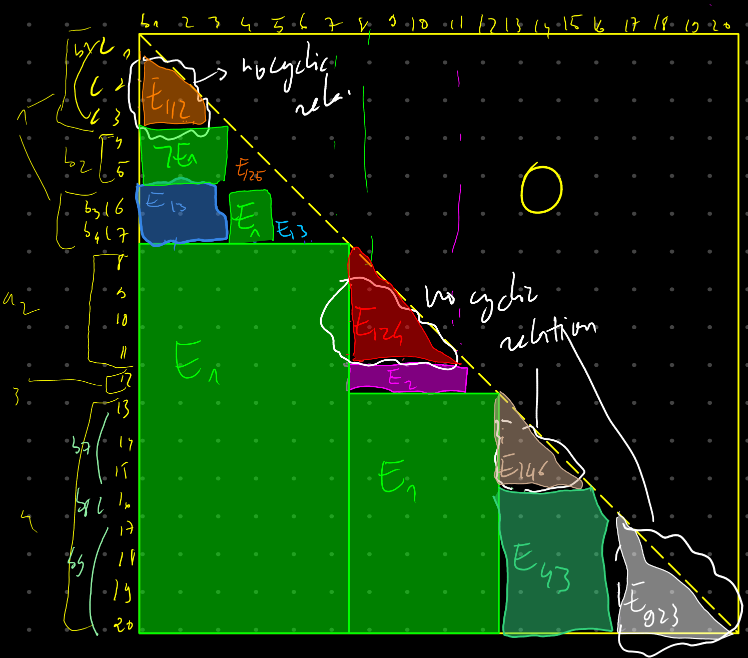
for example:

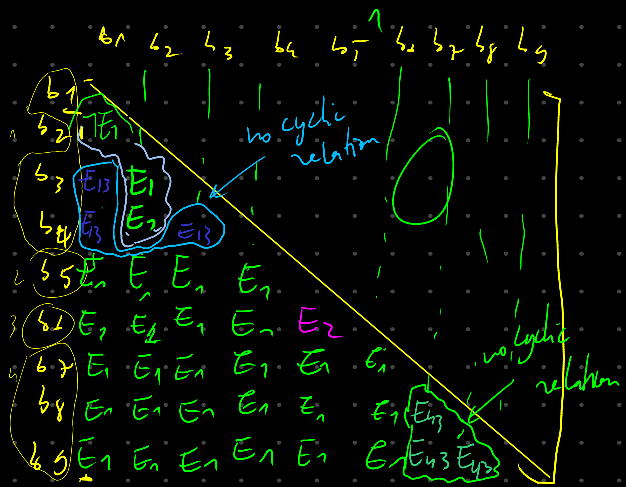
$$\varphi\begin{pmatrix} E_{41} \\ \shortparallel \quad \shortparallel \\ E_{42}\, E_{43} \end{pmatrix} = 1$$



having just 1s, don't lead to cyclic relations

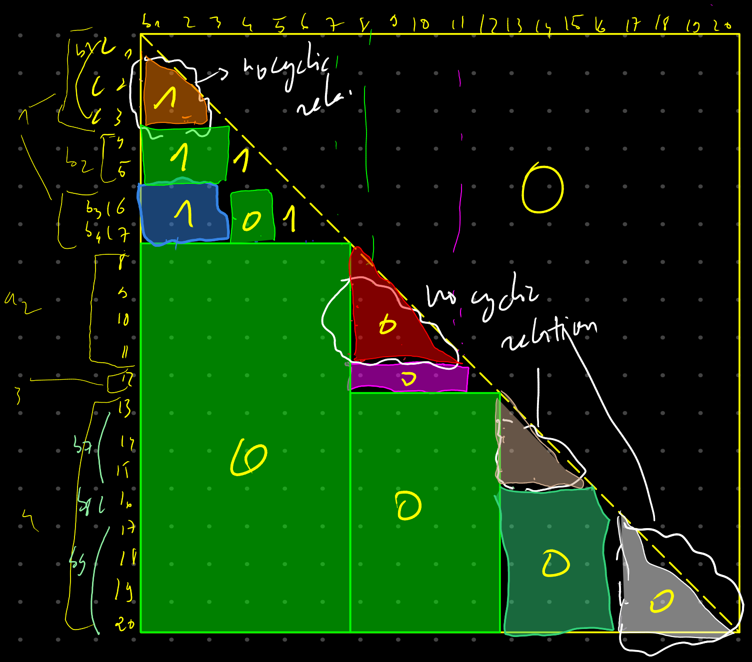
thus, we can reduce the equivalent class to:

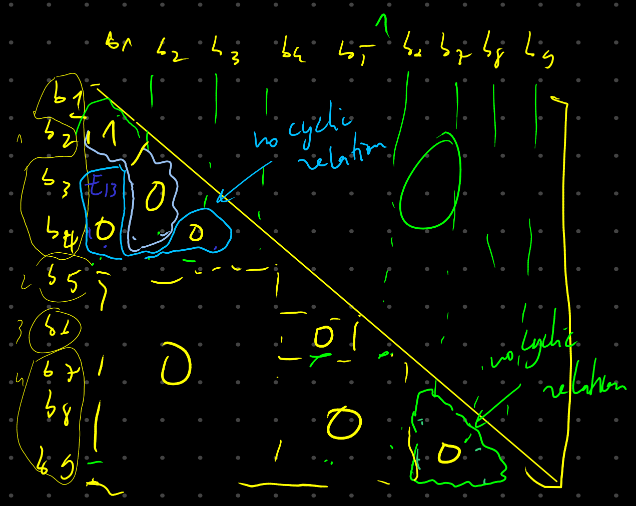


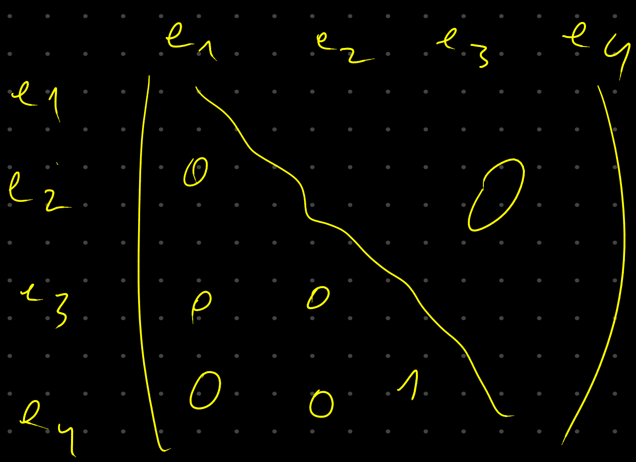


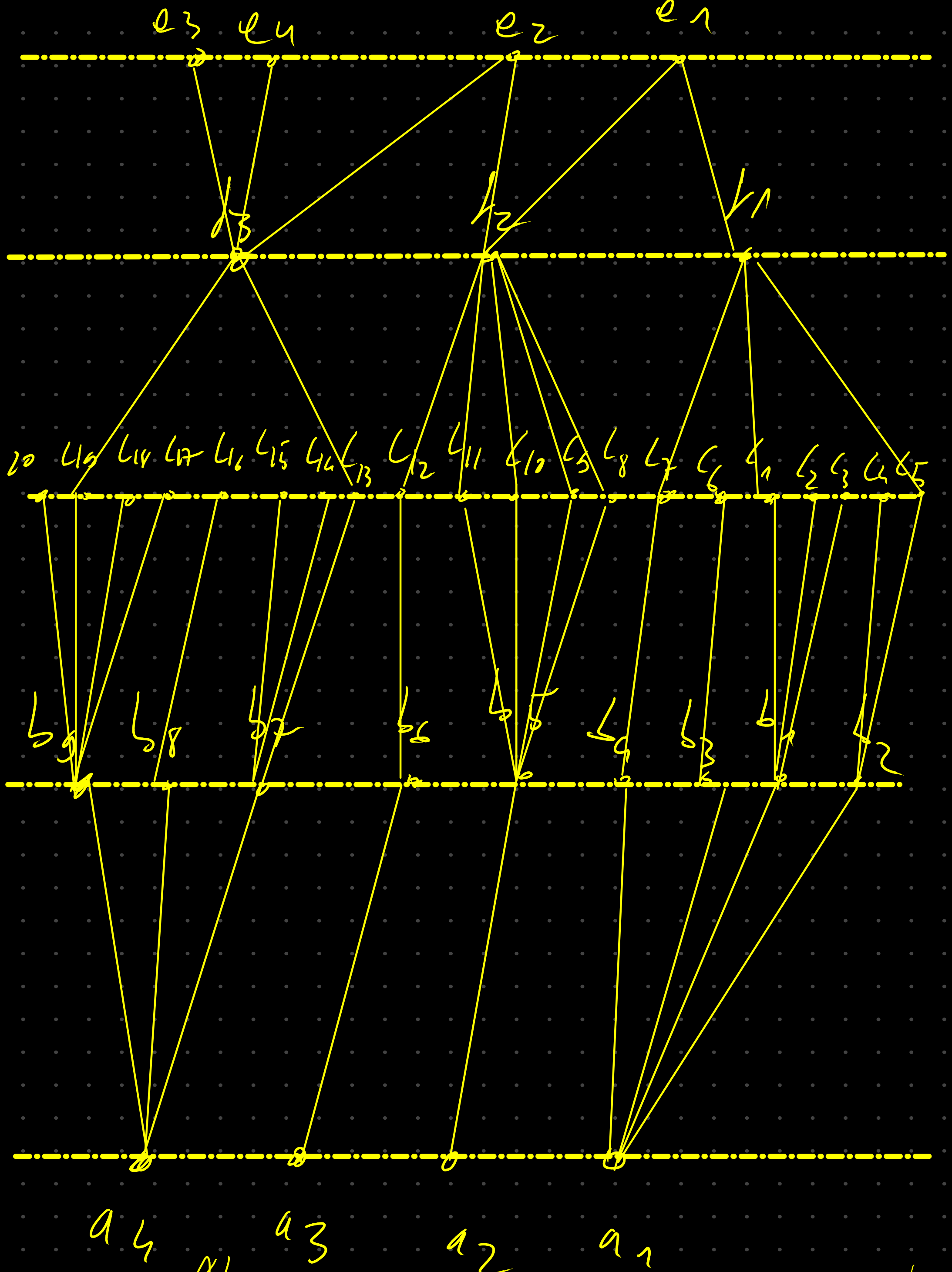
Arbitrary assignment of the equivalent classes that's left:

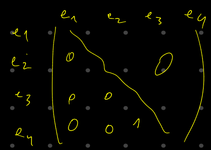
E_1	E_2	E_{13}	E_{43}	E_{112}	E_{124}	E_{125}	E_{146}	E_{923}	E_0
0	0	0	0	1	0	1	0	0	1

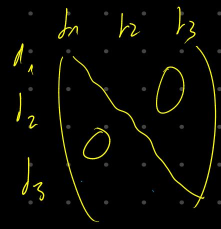


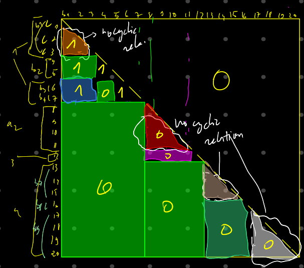


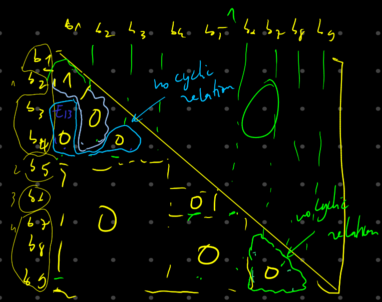








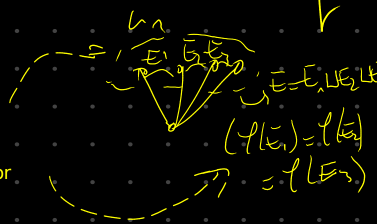




Algo :

group the vertices that are the disjointed vertices of adjacent edges coming from level below them.

concatinate the equivalent classes of these vertices into one equivalent class for each group



find vertices that don't have an incoming edge

for a vertex v that doesn't have an incoming edge :
check vertices from the same level that are part of one these groups

concatinate all the equivalent classes of these vertices
into

(before) v (after)

$\varphi(\neg E_{13}) = \varphi(E_{1})$

in the example :

$\neg E_{12} = E_{15} = E_{16} = E_{1}$